

Ex.No.1

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Study of various Network Commands used in Linux and windows

AIM:

Study of various Network Commands used in Linux and windows.

BASIC NETWORKING COMMANDS:

1. `arp -a` : Address resolution protocol. It shows the IP address and MAC address of router.

output:

- gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3

2. `hostname` : displays the name of system.

output: localhost.localdomain

3. `ipconfig` : displays detailed Configuration information about TCP/IP connection including Router, Gateway.

output: enp0s3 : flags = 4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500

inet 10.0.2.0.15 netmask 255.255.255.0 broadcast

10.0.2.255 inet6 fe80::a00:27ff:fe68:9da0

4. `nbtstat -a` : helps solve problems with NetBIOS

output: NBTSTAT [-a RemoteName] [-A IP address]
[-c] [-n] [-r] [-R] [-RR] [-s] [-s] [interval]

-a (adapter status) Lists the remote machine's name table given its name.

-c (cache) Lists NBT's cache of remote [machine] names and their IP addresses.

-n (names) List local NetBIOS names.

5. **netstat** : (network statistics) displays a variety of statistics about a computer's active TCP/IP.

eg: `netstat -r`

output: Kernel IP routing table

Destination	Gateway	Genmask	Flags	MSS	Window
default	- gateway	0.0.0.0	UG	0	0
10.0.2.0	0.0.0.0	255.255.255.0	U	0	0

6. **nslookup** : (name server lookup) is a tool used to perform DNS lookups in Linux.

eg: `nslookup`

> `www.google.com`

output:

Server: 10.101.43.22

Address: 14.101.43.22#53

7. **pathping** : It is combination of PING and Tracert Commands.

output:

usage: `pathping [-g host-list] [-h maximum-hops]`
`[-i address] [-n] [-p period] [-q num-queries]`
`[-w timeout] [-4] [-6] targetname`

8. **ping** : (Packet Internet Groper) Connecting between two nodes.

output:

PING 4.2.2.2 (4.2.2.2) 56(84) bytes of data:

64 bytes from 4.2.2.2: icmp_seq=1 ttl=52

time=580 ms.

64 bytes from 4.2.2.2: icmp_seq=2 ttl=52

time=370 ms.

9. **Route** : used to show / manipulate IP routing table.

output:

Destination	Gateway	Genmask	Flags	Metric	Ref
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0

LINUX NETWORKING COMMANDS :

1. `ip` : shows address information, manipulate routing.
`ip <OPTIONS> <OBJECT> <COMMAND>`

output:

usage: `ip {options} object {command | help}`
`ip [-force] -batch filename`

where object := {address | addresslabel | fd | help | link | ioam}

2. `ifconfig` : used for configuring and trouble shooting networks.

output:

flags=4163 <UP, inet 10.0.2.15 netmask 255.255.255.0
broadcast 10.0.2.255
RX packets 92 bytes 9059 (8.8 KiB)

3. `mtr` : Matt's traceroute combines the functionality of ping and traceroute commands.

`mtr <options> hostname/IP`

output:

	Packets				Pings		
loss%	Snt	Last	Avg	Best	Worst	stdev	
0.0%	97	0.1	0.1	0.1	0.3	0.0	

4. `tcpdump` : used for capturing and displaying packets.

`$ sudo tcpdump`

output: [sudo] password for Karolin : root.

5. `ping` : that verifies IP-level connectivity to another TCP/IP computer by sending ICMP.

`$ ping google.com`

output:

64 bytes from pmmaa-ba-in-f14-1e100.net
(142.251.221.206) : icmp_seq=1 ttl=115

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Result :

Thus the various network commands used in Linux and windows was executed successfully.